Part 1 What is life?

The core activities in this unit total about 24 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	**Living or non-living?**	ACSSU150	Engage & Evaluate Digital Short	Students explore a variety of stimulus material to develop an understanding of the terms living and non-living.	Optional
1.2	**Characteristics of life**	ACSSU150 ACSHE135 ACSHE136	Explore, Explain & Elaborate Hands-on & Digital Long	Students observe sultanas in carbonated water to simulate a possible new life form and use a second simulation to explore movement as a possible characteristic of life. Students are then introduced to the more formalised characteristics of life and investigate ambiguous situations of life to consolidate their understanding of the term living.	Core
1.3	**Plant life**	ACSSU150 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS148	Explore & Explain Hands-on Long	Students grow broad bean seeds and record observations of their growth throughout the unit. Experimental design and data collecting skills are revised, leading later to **Activity 5.8** where students will be reporting conclusions.	Core
1.4	**Is Sammy alive?**	ACSSU150 ACSHE135 ACSHE136	Elaborate Digital Short	This activity is based on an animation of the life of Sammy, whose body parts have progressively been replaced. Students participate in group discussions to decide at what stage Sammy is no longer alive.	Core

Part 2 What are cells?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	**Cell theory**	ACSSU149 ACSHE134 ACSHE226 ACSHE136	Explore & Explain Classroom & Digital Medium	Students use written material and digital interactives to study the historical and technical development of microscopes and learn about the capabilities of each microscope. In the Find out more students see detailed images of insects, diatoms, blood cells and wood taken with a scanning electron microscope.	Core
2.2	**How small can things be?**	ACSIS141 ACSIS144	Explore & Explain Classroom & Digital Long	Students learn to use and convert units as they explore the concepts of size, measurement and scale diagrams; focusing on units used for microscopic objects. An animation helps visualise the relative size of objects and the appropriate units to use in measurement	Core
2.3	**Making little things bigger**	ACSSU149 ACSIS141	Explore, Explain & Evaluate Hands-on & Digital Medium	Students learn the parts and functions of the microscope and how to set up and focus prepared slides.	Core
2.4	**Observing cells**	ACSSU149 ACSHE134	Explore & Explain Classroom Short	Students use micrographs and scientific drawings of cells to familiarise themselves with basic cell parts.	Core
2.5	**Microscope skills**	ACSIS141	Explore & Explain Hands-on Medium	Students further explore the use of the microscope and learn about the inversion of images and resolution. They learn how to make measurements using minigrids.	Optional
2.6	**Looking at living cells**	ACSSU149 ACSSU150 ACSIS141	Explore & Explain Hands-on & Digital Medium	Students follow instructions to prepare a wet mount of onion skin cells. They focus under low and high magnification, before staining the cells and comparing the effect on the image seen.	Core

Part 2 What are cells? (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.7	**Drawing cells**	ACSIS144	Explain Classroom Medium	The class will view a variety of biological diagrams drawn by scientists and students. Students will learn about drawing biological diagrams and draw labelled cells from photomicrographs.	Core

Part 3 What goes on inside a cell?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	**What's inside a cell?**	ACSSU149 ACSIS144	Explore, Explain & Elaborate Digital Short	Students view stylised cell diagrams to explore the classification of cells and use their findings to classify cells.	Core
3.2	**What makes cells function?**	ACSSU149	Explore, Explain & Elaborate Digital Medium	A **jigsaw** activity in which groups explore the structure and function of different cell organelles and regroup to share findings. The Notebook activity introduces single-celled organisms and invites students to apply their knowledge to new situations.	Core
3.3	**Yeast metabolism**	ACSSU149 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore & Explain Hands-on Long	In this activity students will explore the metabolism of yeast by planning and carrying out a controlled experiment to investigate the effect of temperature on cell activity, then graphing the results. They will also use a microscope to observe yeast budding.	Core
3.4	**Cell division**	ACSSU149 ACSSU150	Explore & Explain Digital Short	Students use digital media to learn about binary fission and mitosis as methods of cell division involved in growth and reproduction.	Core
3.5	**Why are cells so small?**	ACSSU149 ACSIS140 ACSIS141	Explore & Explain Hands-on Medium	Students carry out an investigation using agar blocks to represent different sized cells in order to study the efficiency of diffusion in different sized cells.	Core
3.6	**Practical skills in biology**		Evaluate Hands-on & Classroom Medium	A practical skills test for Units 1 - 3 (editable version) and marking scheme are available to teachers by email request from *Science by Doing* (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - *Science by Doing* provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. *Science by Doing* encourages teachers to modify the items to suit individual teaching programs.	Optional

Part 4 How can little cells get together to make big things?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	**Life in water**	ACSSU149 ACSSU150	Engage & Explore Hands-on Medium	Students observe organisms in pond water using microscopes.	Optional
4.2	**Cells specialise**	ACSSU150	Explore, Explain & Elaborate Hands-on & Digital Medium	Students investigate tissues using prepared slides, wet mounts, and virtual microscope slides.	Core
4.3	**Putting it all together**	ACSSU150	Explore, Explain & Elaborate Digital Short	The concept of cell-tissue-organ-system-organism is explored using simulations of human and flowering plant systems.	Core
4.4	**Organs and systems**	ACSSU150	Explore & Explain Digital Medium	Students ponder the importance of body systems and whether the body can survive without one of them. They investigate the structure and function of the digestive system. The Find out more section introduces the skin as a body organ, discussing its structure and function, skin burns and skin cancer.	Core
4.5	**New skin**	ACSSU150 ACSHE134 ACSHE226 ACSHE136	Elaborate Classroom & Digital Medium	Students study the life and work of 2005 Australian of the Year Professor Fiona Wood, a skin burns surgeon, using several media accounts.	Core
4.6	**Let's look inside**	ACSSU150	Explore & Explain Hands-on Medium	Students dissect a squid as an example of a functioning organism.	Optional

Part 4 How can little cells get together to make big things? (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.7	**Plant plumbing**	ACSSU150	Explore & Explain Hands-on Medium	Students do experiments to study a plant vascular system and the structure and role of stomata.	Optional
4.8	**How does a multicellular organism function?**	ACSSU150	Evaluate Classroom Short	A formative assessment where students use packs of cards to link the components (cell-tissue-organ-system-organism) of the circulatory, digestive and excretory systems.	Optional

Part 5 Cells die, but life goes on!

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	**Is death a part of life?**	ACSSU149	Engage, Explore, Explain & Elaborate Digital Medium	Students investigate the cycle of life and death of cells by interpreting an animation of cells in intestinal villi.	Optional
5.2	**A scientific breakthrough!**	ACSHE134 ACSHE135 ACSHE136 ACSIS139	Explain & Elaborate Digital Medium	Students investigate the life and work of Professor Barry Marshall by listening to an interview and exploring other media sources. They study how he made the link between the bacterium Helicobacter and stomach ulcers and cancer.	Core
5.3	**Why the difference?**	ACSSU150	Explore, Explain & Elaborate Classroom Short	Students compare and contrast organ systems of animals from different niches. They explore the relationships between teeth and diet, and skeletal systems and lifestyle.	Core
5.4	**Reproductive strategies in animals**	ACSSU150	Explore, Explain & Elaborate Classroom & Digital Long	Students analyse written and visual information to make comparisons between animals with different reproductive strategies. They relate the reproductive strategies to the animal's niche and classification.	Core
5.5	**Plants reproduce too!**	ACSSU150	Explore, Explain, Elaborate & Evaluate Hands-on & Digital Long	Students study sexual reproduction in plants using digital media and hands-on experiences to learn about the structure of flowers and their function in pollination and fertilization. Students evaluate their knowledge of flower parts (**formative assessment**) in a click-and-drag exercise.	Core

Part 5 Cells die, but life goes on! (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.6	**But there's another way!**	ACSSU150	Explore & Explain Hands-on & Classroom Medium	Students investigate methods of asexual reproduction in plants. This is followed by discussion of the advantages and disadvantages of reproductive methods.	Core
5.7	**Presenting and interpreting experimental results**	ACSIS144 ACSIS145	Explain & Elaborate Classroom Medium	Students consolidate their skills of tabulation, graphing and drawing valid conclusions through guided activities.	Core
5.8	**So how did it grow?**	ACSSU150 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS148	Explain, Elaborate & Evaluate Classroom Medium	Students correlate the information collected during the growth of their bean and write a scientific report on their findings. This can be used as a **summative assessment**.	Core
5.9	**Sample test**		Evaluate Classroom Medium	A sample summative test for Units 1 - 5 (editable version) and marking scheme are available to teachers by email request from *Science by Doing* (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - *Science by Doing* provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. *Science by Doing* encourages teachers to modify the items to suit individual teaching programs.	Optional

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